Quantum Fault Tolerance Meets Toom's Contours

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1 Toric Encoded Circuits

[COMMENT] Choose one of the following definitions for the base graph G_o. Pitori Berman and Peter Gacs chose the second, namely the projected version, yet if I had to guess, I would say that the first gives an advantage. Furthermore, We might want to fold computation stages, such that we have GATE → NOISE → DECODING, and after each such cycle freeze the state of the graphs. Finally, we might want that the gates will be SIZE-invariant, like rotating the Torus.

Definition 1.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \dots, R_k$, and each of the registers is encoded by $\mathcal{C}$.
2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, .., D_1 P_1 \Lambda_1$$

Definition 1.2 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \left\{(v, t) : t \in [d], v \in \bigcup_i \mathcal{G}_i\right\}$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned} E_o := \big\{ &\{(v,t),(u,t')\} : \text{ if} \\ &\text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ &\text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1}+1 \big\} \end{aligned}$$

Definition 1.3 (Base graph G_o)**.** Let $C = \{u_i\}_{i=1}^{n}$ be a quantum circuit, at depth d, with registers $R_1, \ldots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}$ the Toric complex that induces the codes, and suppose that the locations of the gates in C are given in registers' coordinate systems. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in $\mathcal{G}$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}$, with the addition of edges of the following form: For two vertices at preceding times t and $t+1$, let's denote them by (v_1, t) and $(v_2, t+1)$, for which there exist registers R_1', R_2', and faces $f_1, f_2 \in \mathcal{G}$, such that $v_1 \in f_1, v_2 \in f_2$ and C contains a gate that acts non-trivially on the locations $(t, f_1, \mathbf{1}_{R_1'})$ and $(t, f_2, \mathbf{1}_{R_2'})$ at time t, are also connected by an edge. Namely:

$$\begin{aligned} E_o := \big\{ &\{(v,t),(u,t')\} : \text{ if} \\ &\text{either} \{v,u\} \in \mathcal{G} \text{ and } t' = t, t+1 \\ &\text{or there exists } u_i = (g_i, l_i) \in C, f_1, f_2 \in \mathcal{G}, \text{ and } R_1' R_2' \\ &\quad \text{such } v \in f_1, u \in f_2, t = l_{i,1}, t' = l_{i,1}+1 \\ &\quad \text{and } (f_1, \mathbf{1}_{R_1'}), (f_2, \mathbf{1}_{R_2'}) \in l_{i,2} \big\} \end{aligned}$$

Definition 1.4 (Time Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$. We define the history graph $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are taken to be the vertices of G_o, namely tuples $(v,t) \in V_o$, that satisfy that v belongs to a face associated with a qubit, which, at time t, belongs to the deviation. Namely:

$$\big\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a qubit } f \in C \text{ such } v \in f \in \mathcal{D}_t \big\}$$

2. The edges $E_{\mathcal{H}}$ can be parttion to two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, the edges in E_o restricted to $V_{\mathcal{H}}$. Namely:

$$E_{\mathcal{H}} = \left\{ e = \{u_t, v_{t'}\} \in E_o : u_t, v_{t'} \in V_{\mathcal{H}} \right\}$$

By definition, the construction $G_{\mathcal{H}}$ is a time-graph.

Definition 1.5 (**Size**()-Function)**.**

$$\mathbf{Size}(S) = \sum_{i}^{d+1} \max_{v \in S} L_i(v)$$

Claim 1.6. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^{k} S_i) \leq \sum_{i=1}^{k} \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cup R_2) &= \sum_{i}^{d+1} \max a_{i,1}, a_{i,2} = \sum_{i}^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_{i}^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

□

Definition 1.7 (Size Invariant Action)**.** [**COMMENT**] Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z, it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 1.8. The sweep rule is Size ξ-shrinking.

Proof. □

[**COMMENT**] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time $= 3t$, in that we apply an invariant action so.